Sampling for Region-Aggregated Spatial Scan Statistics

Foad Namjoo[1] · Drew McClelland[2] · Michael Matheny[3] · Jeff M. Phillips[1]

[1] University of Utah [2] Taptap Send [3] Meta

ACM SIGSPATIAL
Riverside, California
November 5, 2026

What are spatial scan statistics?

What are spatial scan statistics?

Today

Nov 5, 2026



Confirmed COVID-19 cases, Johns Hopkins CSSE · 1 red dot ≈ 100 cases

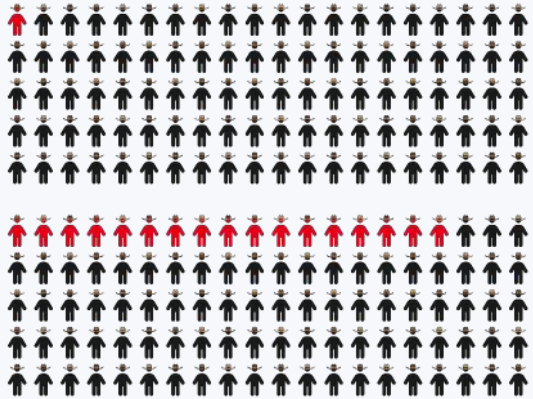
How is a window scored?

In this window:

1.3% of the population lives here

expected **1.3%** of cases

found **17%** of cases



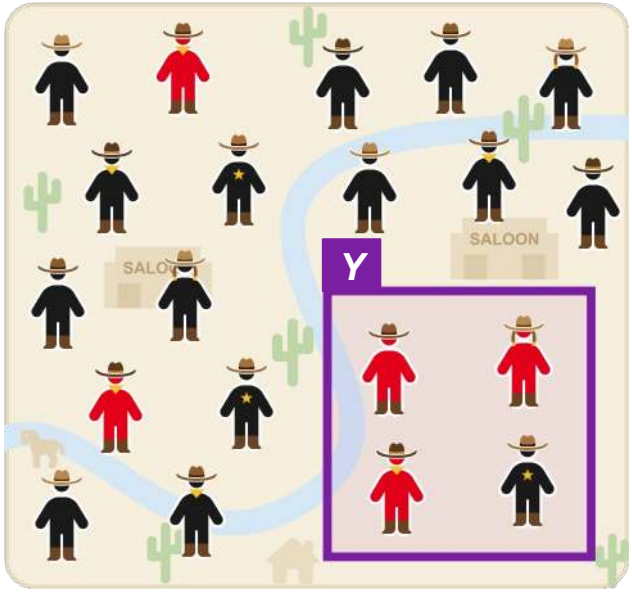
each figure = 1% of all cases

13× more than expected → **high score**



expected = the same case rate everywhere

The Kulldorff score: count the people and the cases



- 20 people
- 5 of them are cases

$$\phi(Y) = m(Y)\log\frac{m(Y)}{b(Y)} + (1-m(Y))\log\frac{1-m(Y)}{1-b(Y)}$$

Y a window on the map

φ(Y) its score: higher = more unusual

b(Y) **people:** 4 of the 20 are in Y



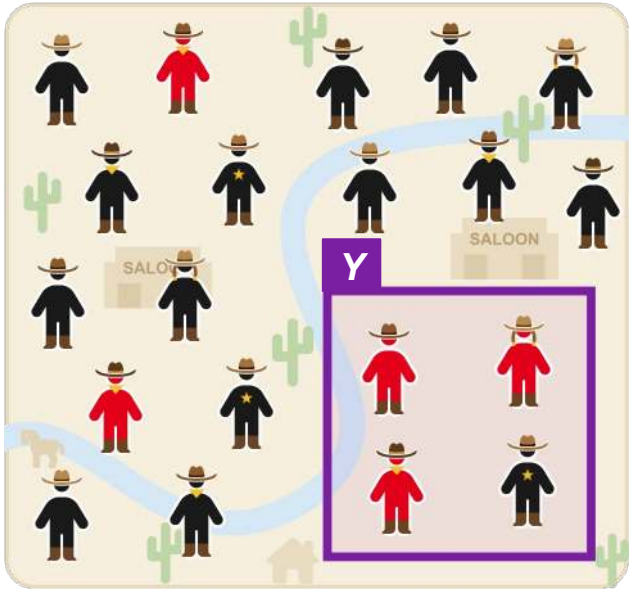
0.2

m(Y) **cases:** 3 of the 5 are in Y



0.6

The Kulldorff score: count the people and the cases



b = 4 / 20 = 0.2 ***m*** = 3 / 5 = 0.6

$$\phi(Y) = m(Y)\log\frac{m(Y)}{b(Y)} + (1-m(Y))\log\frac{1-m(Y)}{1-b(Y)}$$

Inside Y

expected 1

found 3

0.6 × log(**0.6** / **0.2**) = **0.66**

Outside Y

expected 4

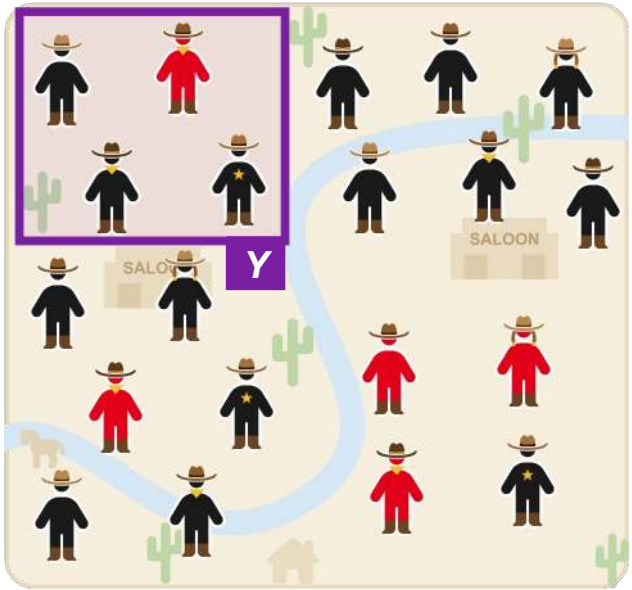
found 2

0.4 × log(**0.4** / **0.8**) = **−0.28**

φ(Y) = 0.66 − 0.28 = **0.38** **hotspot**

expected cases = the area's share of the people × all 5 cases

The Kulldorff score: count the people and the cases



b = 4 / 20 = 0.2 ***m*** = 1 / 5 = 0.2

$$\phi(Y) = m(Y)\log\frac{m(Y)}{b(Y)} + (1-m(Y))\log\frac{1-m(Y)}{1-b(Y)}$$

Inside Y

expected 1

found 1

0.2 × log(**0.2** / **0.2**) = **0**

Outside Y

expected 4

found 4

0.8 × log(**0.8** / **0.8**) = **0**

φ(Y) = 0 + 0 = **0** **no hotspot**

The scan keeps the window with the highest ***φ***. On the globe: ***b*** = 1.3%, ***m*** = 17%, ***φ*** = **0.30**

expected cases = the area's share of the people × all 5 cases

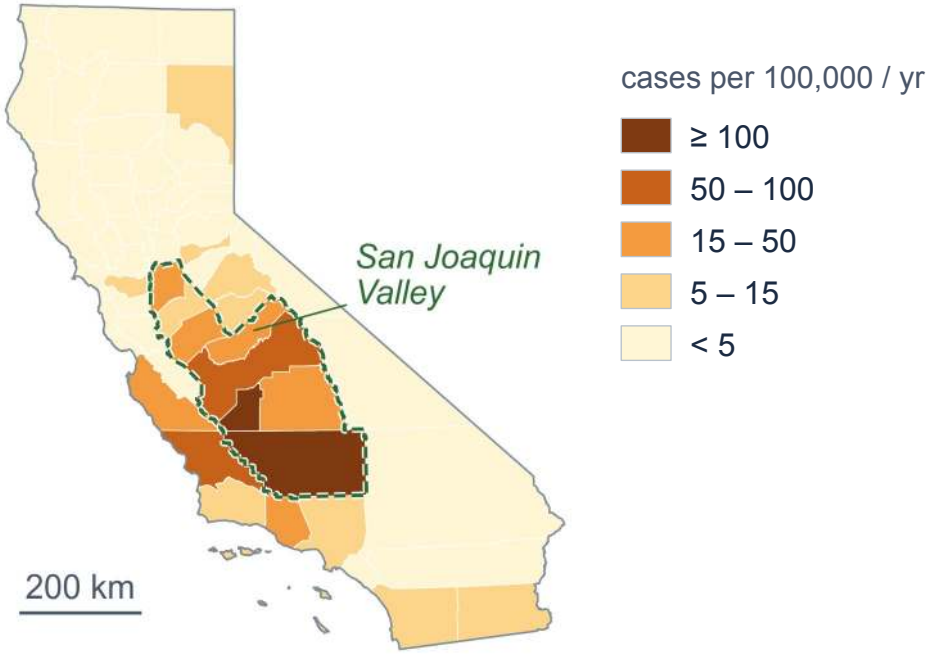
The common tools work on points

Input: each person • and each case • is a point

SaTScan	The most widely used. For regions it uses one point, the centroid, which is our main baseline.
pyScan	**This is what we use.** Open source, Python on top of C++. It scans rectangles and disks over points. mmath.dev/pyscan

For privacy, real data come as counts per region

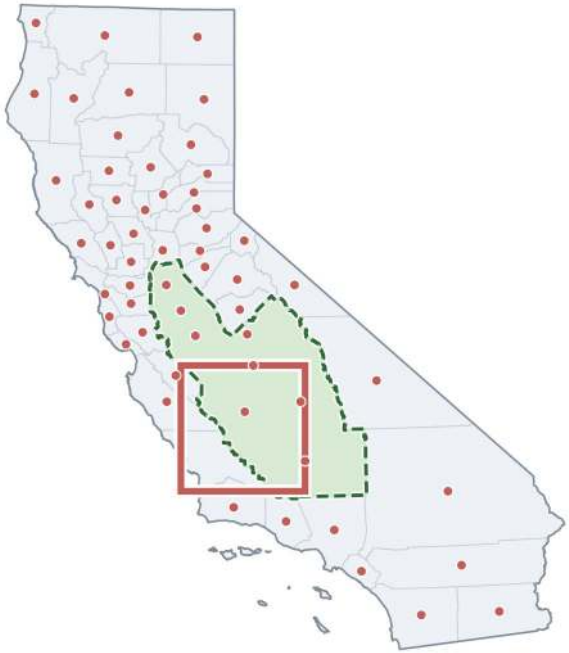
For privacy, real data come as counts per region



Valley Fever cases, California, CDPH 2014–2018

For privacy, real data come as counts per region

Centroid (baseline): 1 point per county

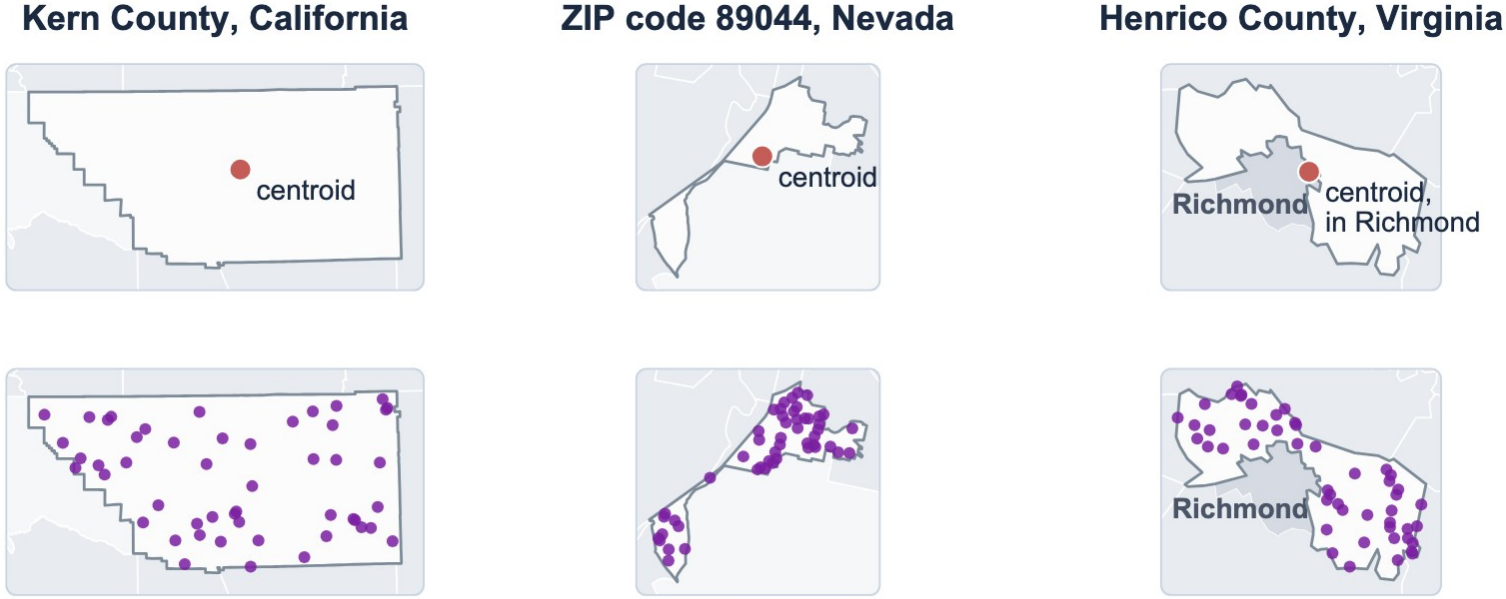


Valley Fever cases, California, CDPH 2014–2018

One point cannot stand for a whole region

Centroid: 1 point

Ours: 50 points



Our method: 50 random points per region

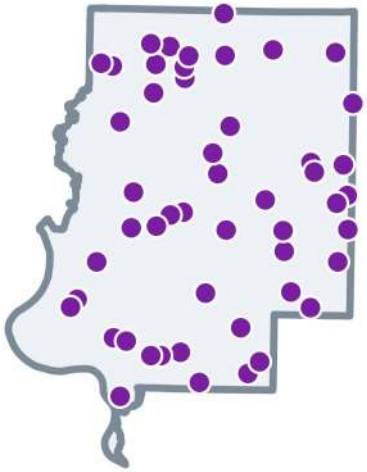


100 cases

10,000 people

split into k = 50 points

each point carries 1/k of the region's cases and people



100 cases / 50 points = 2 cases each point

200 people each

50× more points, but still fast

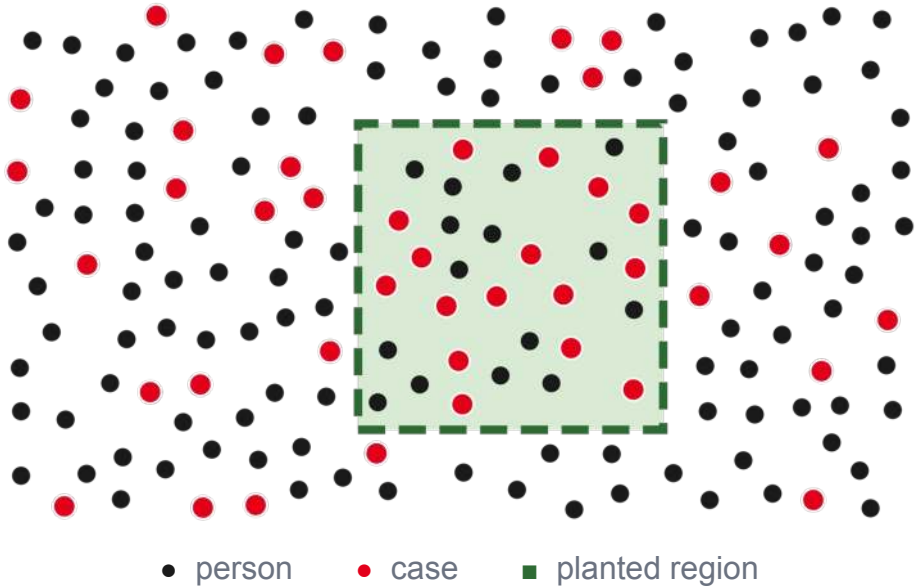
Scan time	1 point	50 points (ours)	Buchin et al.	FlexScan
USA (3,711 regions)	0.29 s	**0.33 s**	114 s	1,109 s
Arkansas (75 counties)	0.21 s	**0.28 s**	25.3 s	8.00 s

pyScan's time is set by its data-adaptive grid, not by the points

At national scale, every county becomes 50 points and the scan finds the hotspot



How we test: we plant a hotspot



$p = 0.5$

inside the box, each person is a case with this probability

$q = 0.2$

outside the box

Signal = $p - q = 0.3$

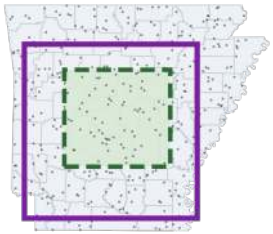
A bigger gap makes the hotspot stand out more.

then the scan runs on the points and returns one window

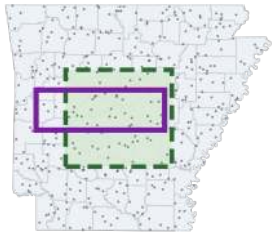
How we test: we score the window the scan finds

Point Jaccard distance $1 - \frac{\text{points in both boxes}}{\text{points in either box}}$

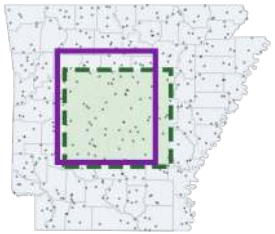
0 = perfect match
1 = no overlap



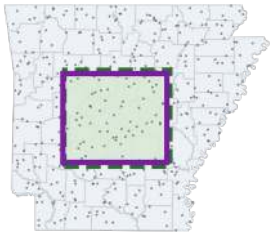
0.67
miss



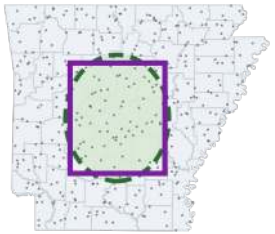
0.37
partial



0.26
partial



0.10
good match



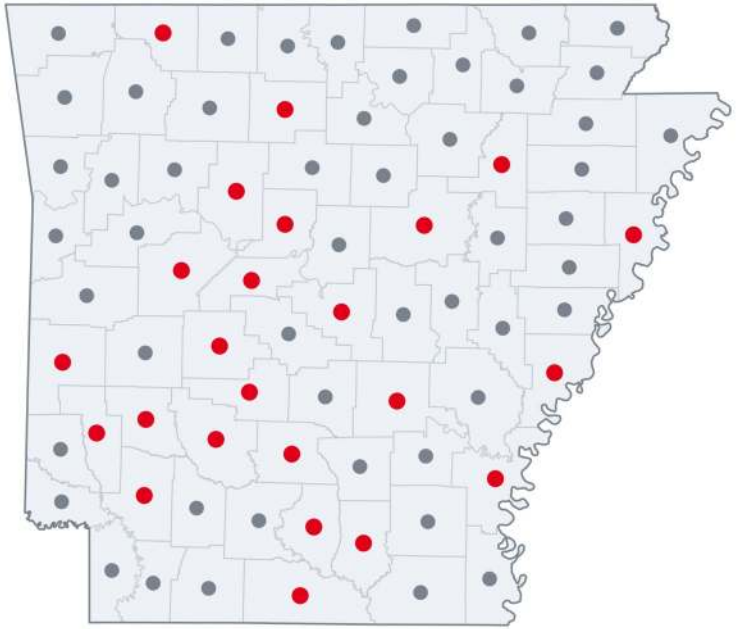
0.16
circle vs rectangle

About 0.2 is as good as we can hope for: the shapes never match exactly

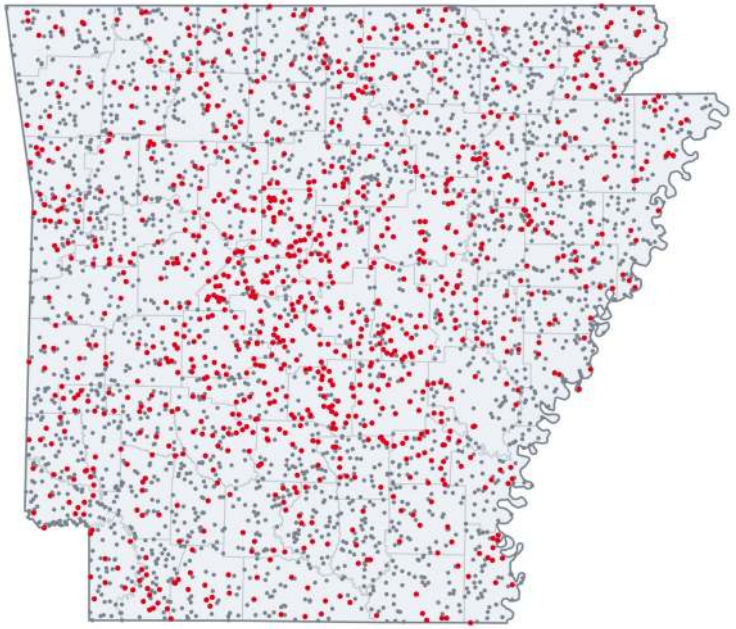
green dashed = planted · purple = found · our method also works with circles · details in the paper

A synthetic test on Arkansas: where is the hotspot?

Centroid (baseline): 1 point per county



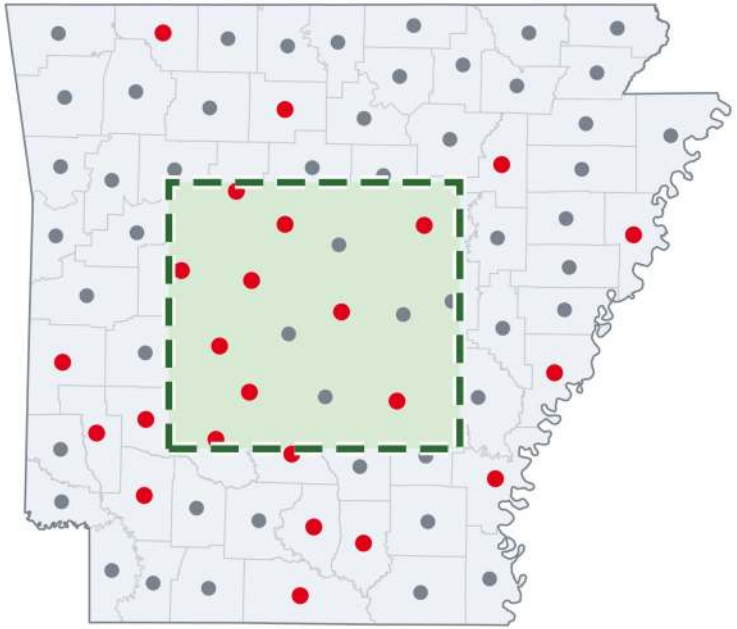
Ours: 50 points per county



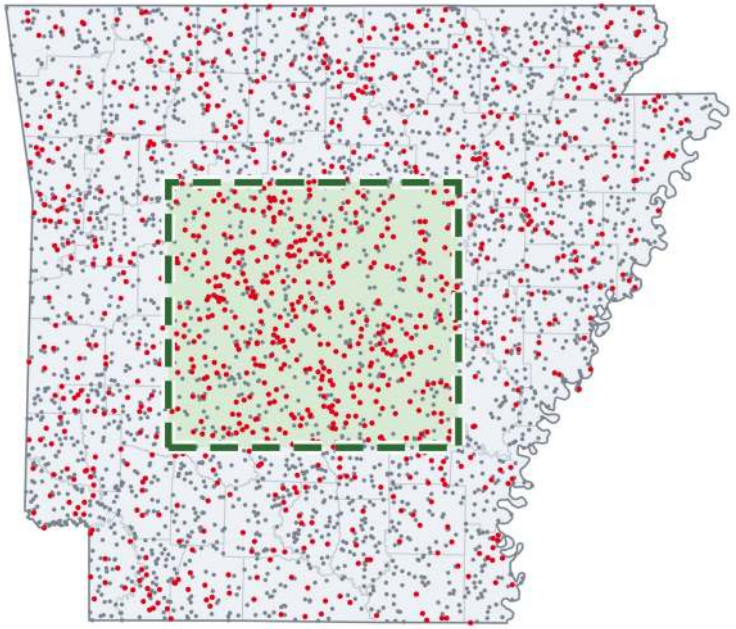
● population ● cases

A synthetic test on Arkansas: where is the hotspot?

Centroid (baseline): 1 point per county



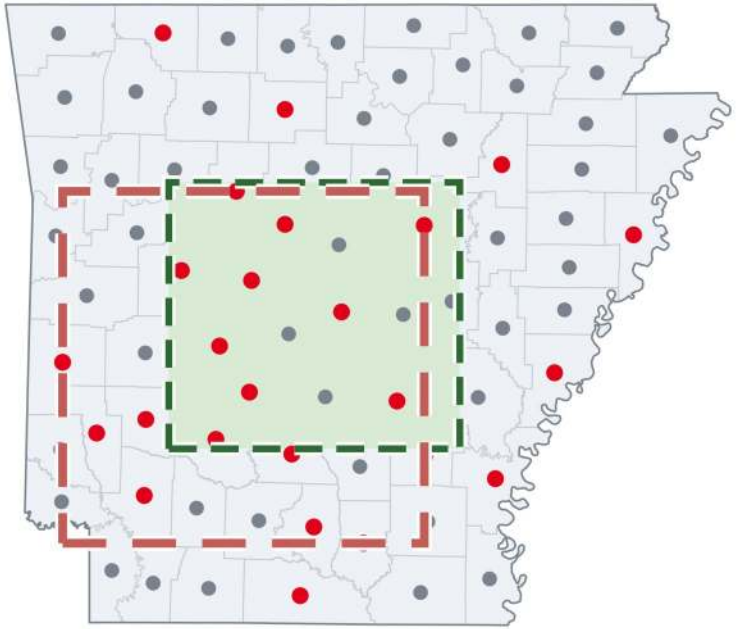
Ours: 50 points per county



● population ● cases ■ planted hotspot: p = 0.5 inside, q = 0.2 outside

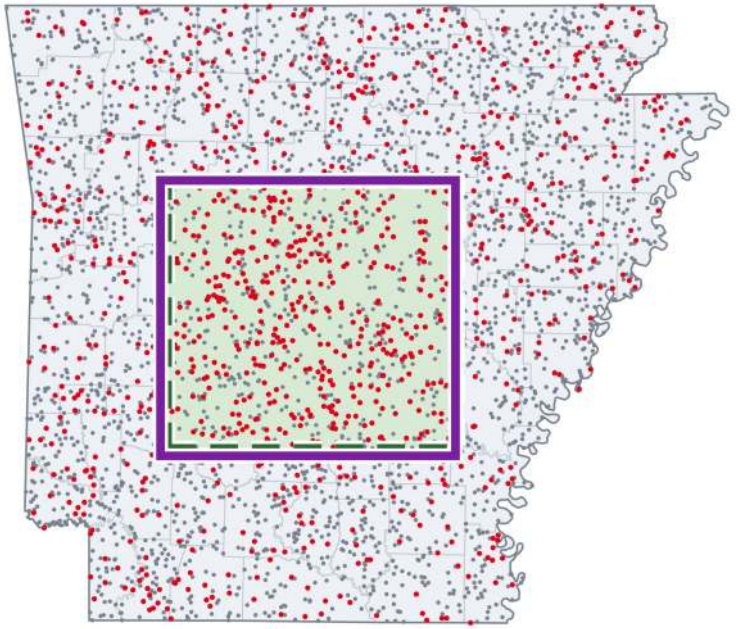
50 points find the hotspot; the centroid does not

Centroid (baseline): 1 point per county



Point Jaccard distance **0.54**

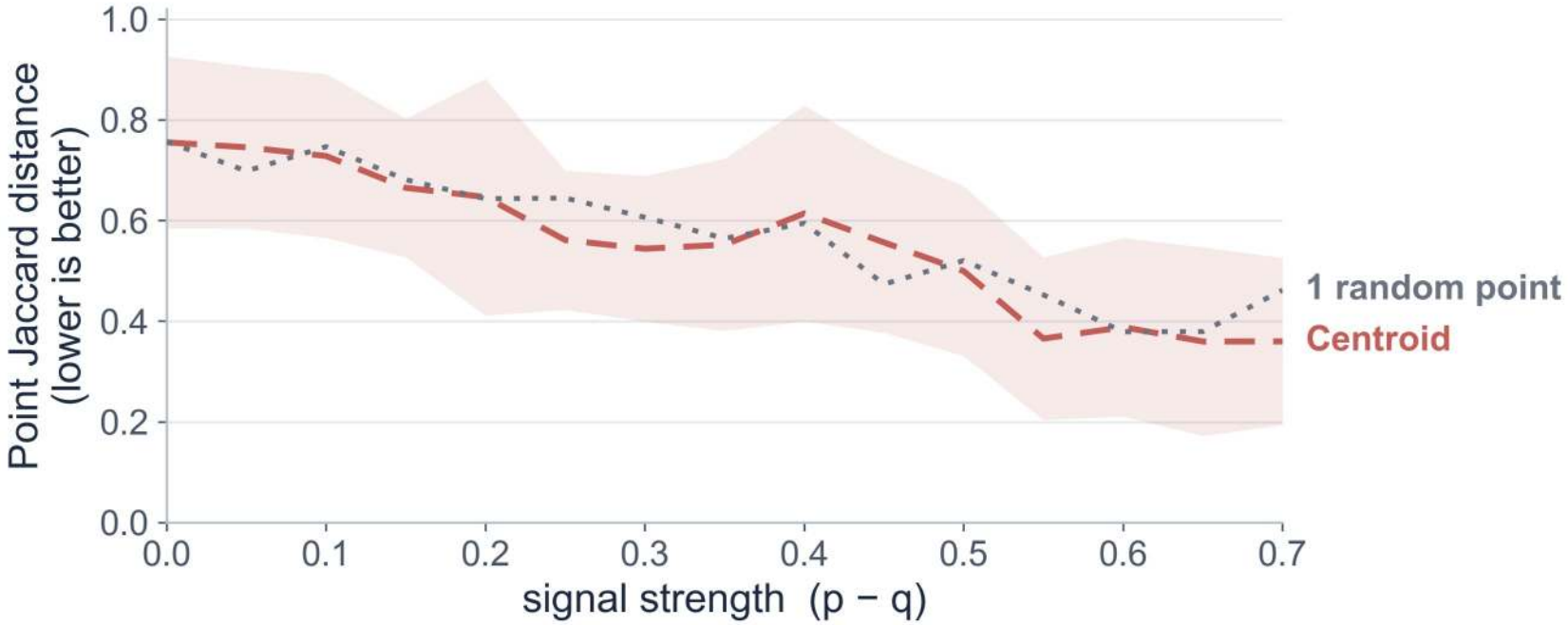
Ours: 50 points per county



Point Jaccard distance **0.08**

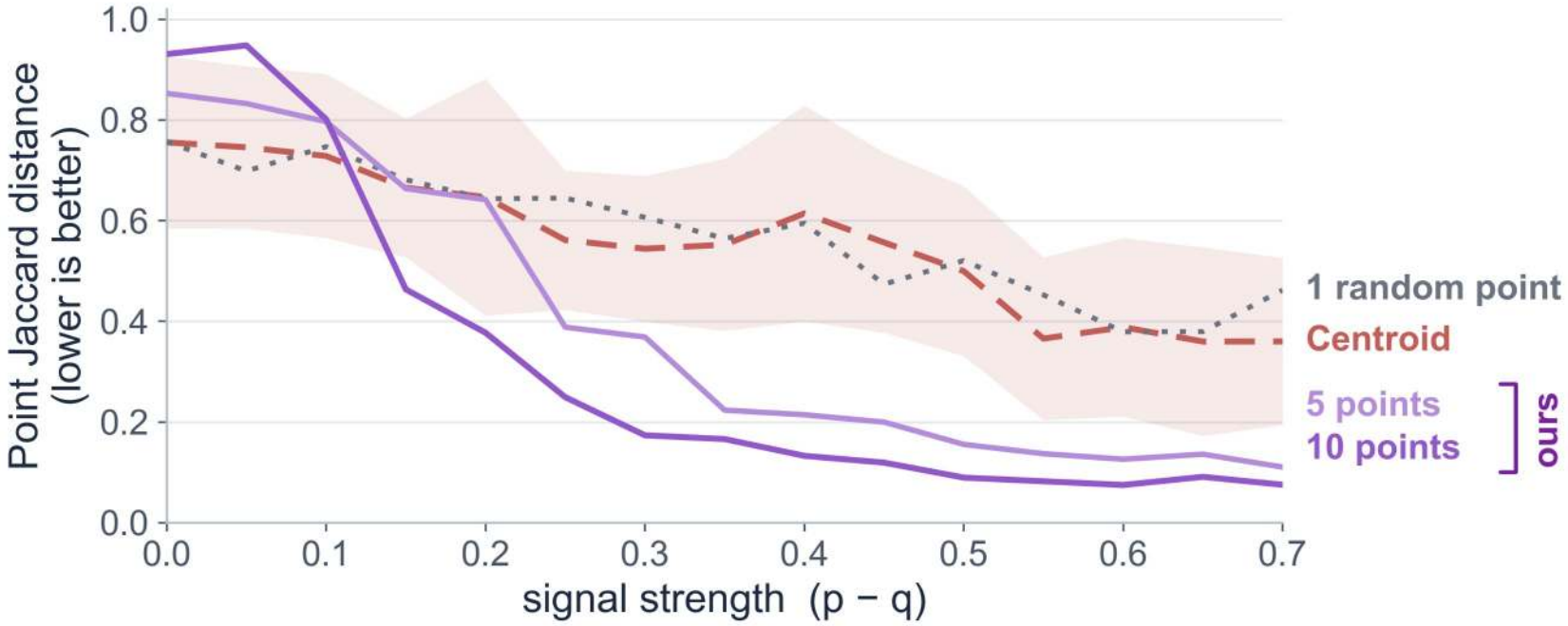
distance counted on 500 fixed points per county · 0 = perfect match, 1 = no overlap

More points, better detection



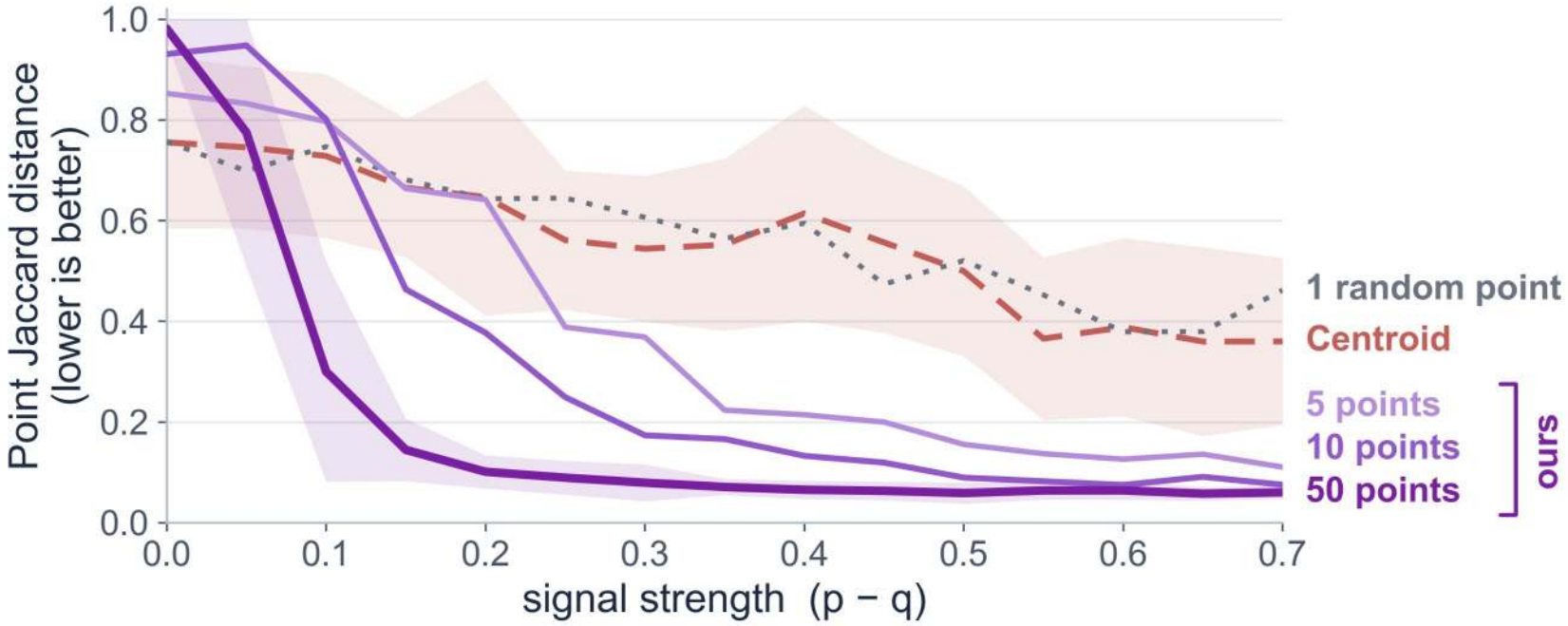
planted region = 30% of Arkansas · q = 0.2 · 20 trials per point · bands = ±1 sd

More points, better detection



planted region = 30% of Arkansas · q = 0.2 · 20 trials per point · bands = ±1 sd

More points, better detection



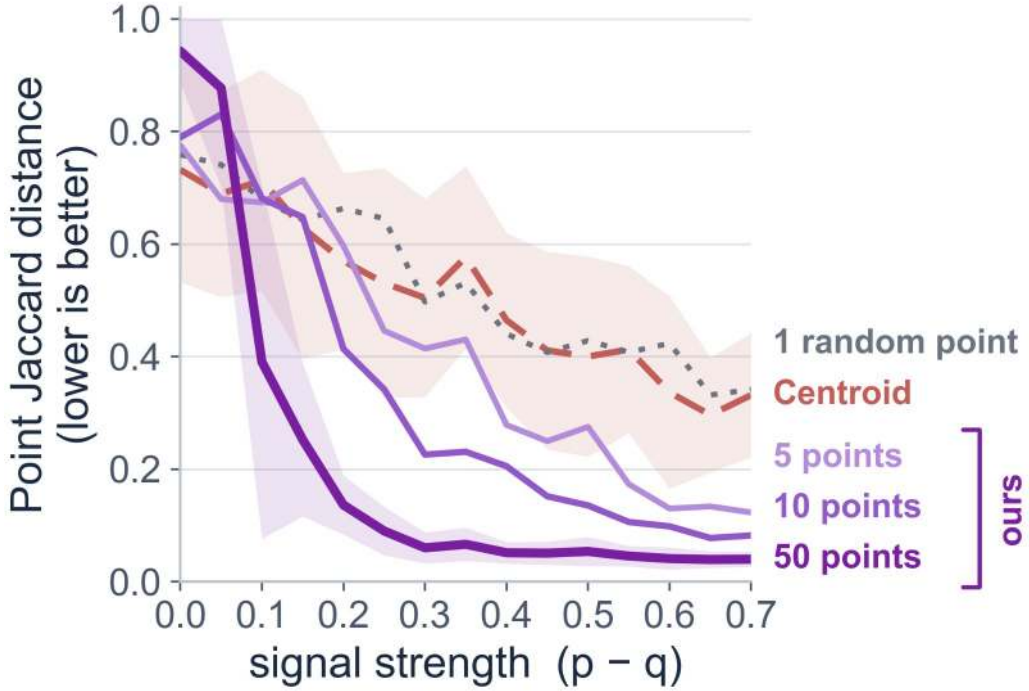
planted region = 30% of Arkansas · q = 0.2 · 20 trials per point · bands = ±1 sd

Sampling wins on all six maps



Utah · 29 counties

big, blocky counties: one centroid stands for a huge area



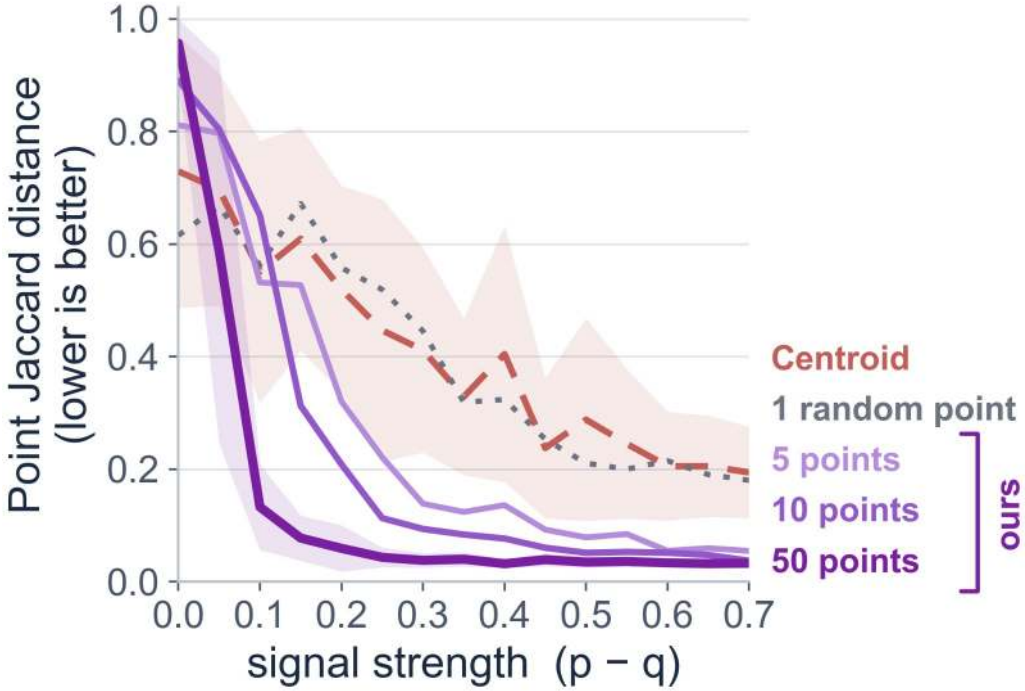
20 trials per point · bands = ±1 sd

Sampling wins on all six maps



California · 69 polygons

long and thin: tiny Bay Area counties next to huge deserts



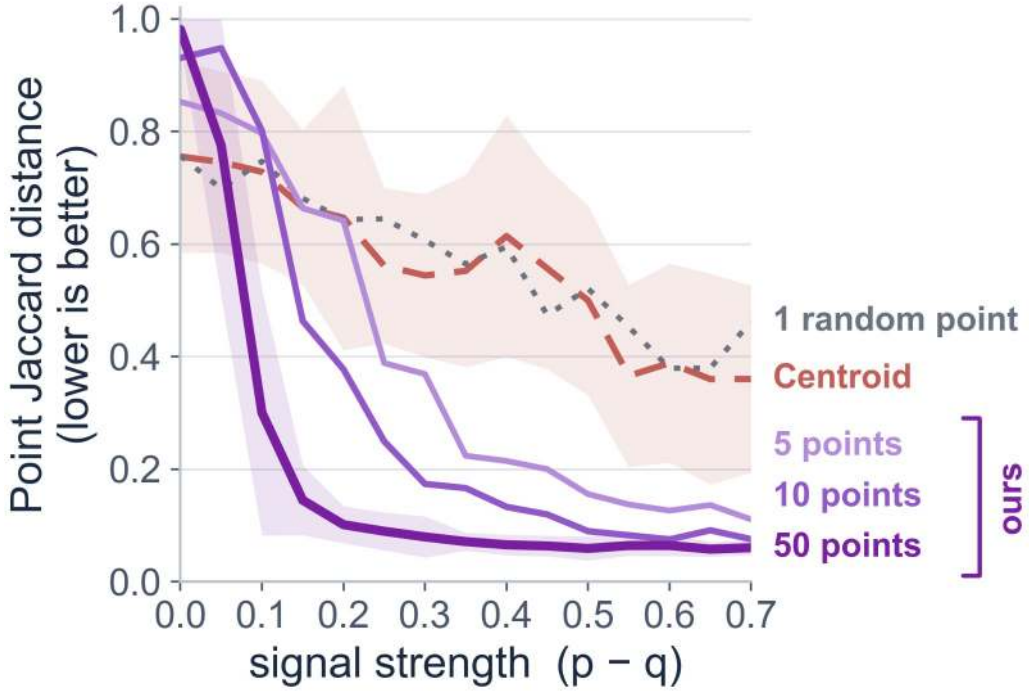
20 trials per point · bands = ±1 sd

Sampling wins on all six maps



Arkansas · 75 counties

mid-sized counties, almost a grid



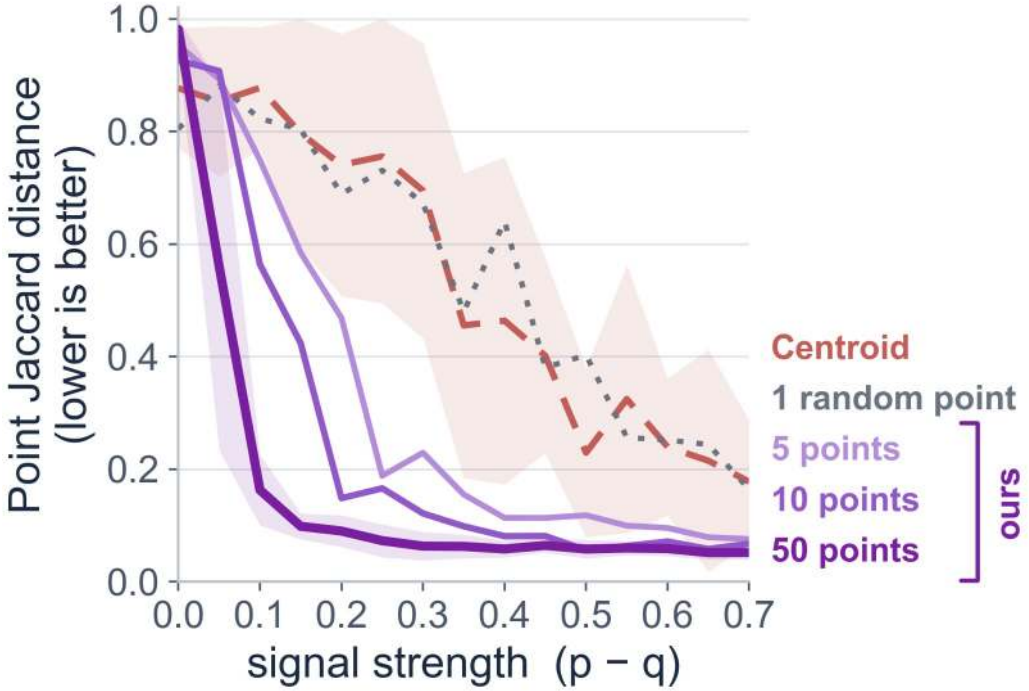
20 trials per point · bands = ±1 sd

Sampling wins on all six maps



Georgia · 159 counties

many small counties: the finest state map



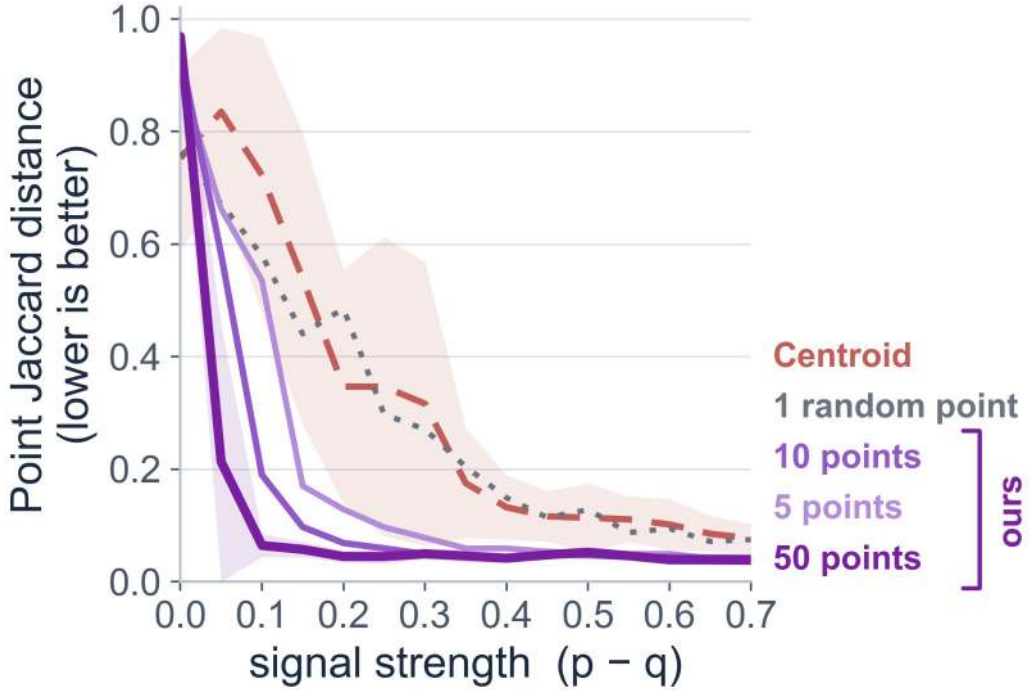
20 trials per point · bands = ±1 sd

Sampling wins on all six maps



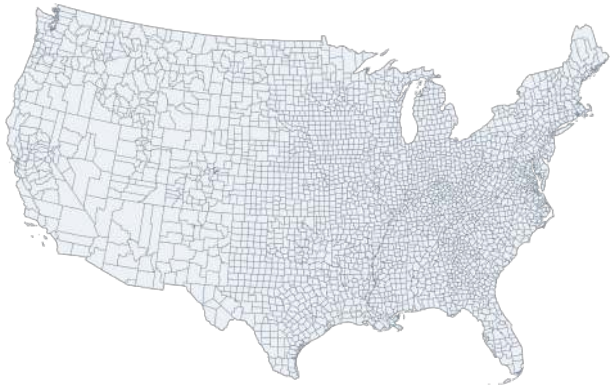
New York City · 263 ZIP codes

tiny, irregular ZIP codes, split by water



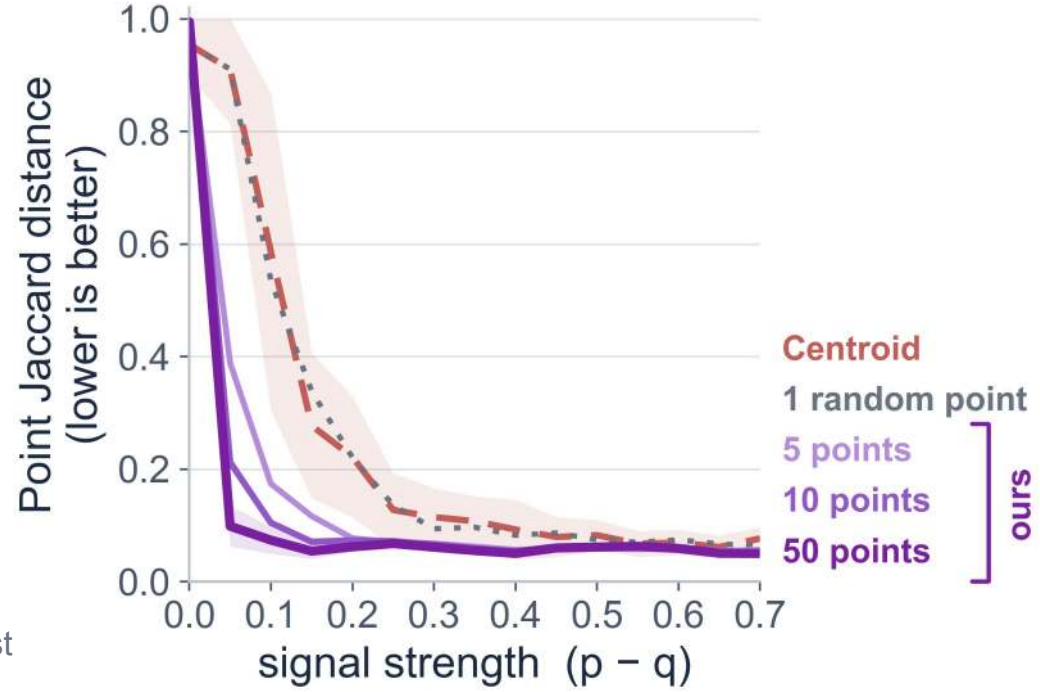
20 trials per point · bands = ±1 sd

Sampling wins on all six maps



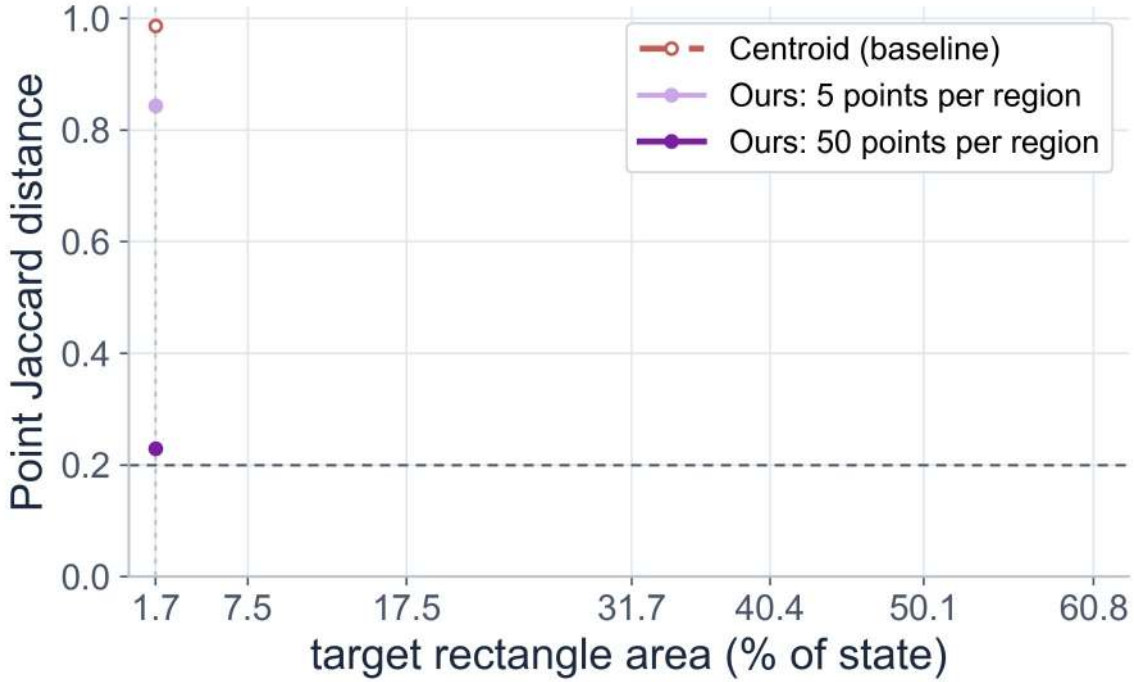
Contiguous US · 3,108 counties

tiny counties in the east, huge ones in the west



20 trials per point · bands = ±1 sd

Georgia: we find regions 15× smaller

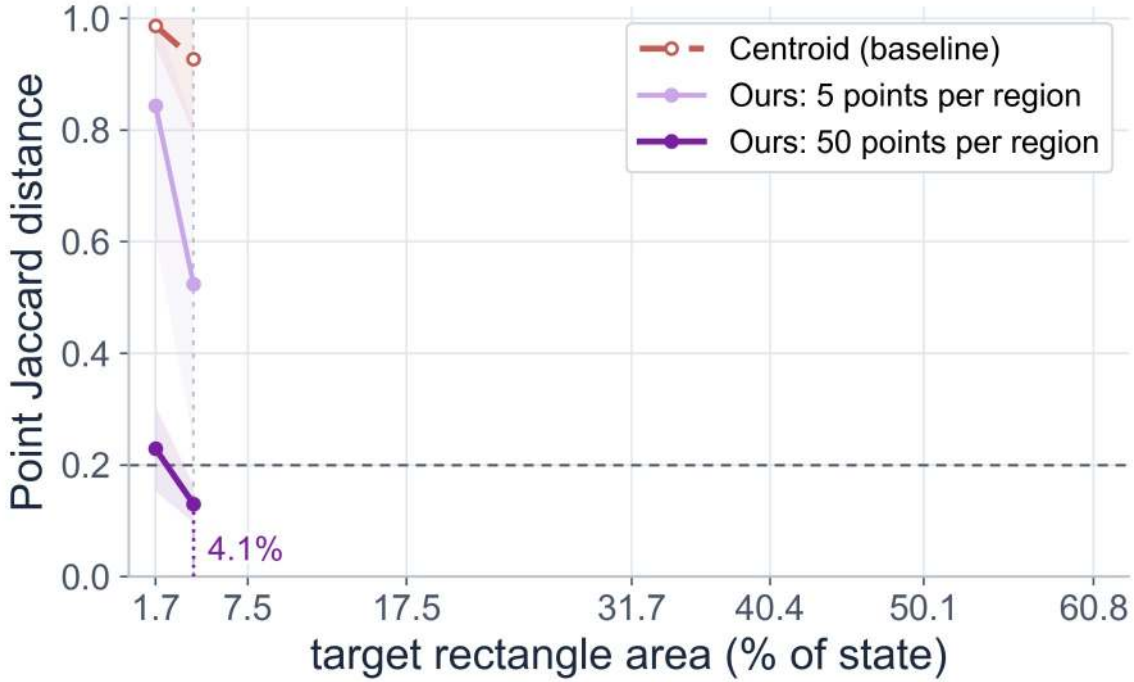




Start tiny. Nobody finds it.

Georgia, 159 counties · p − q = 0.4 · 80 trials per size

Georgia: we find regions 15× smaller

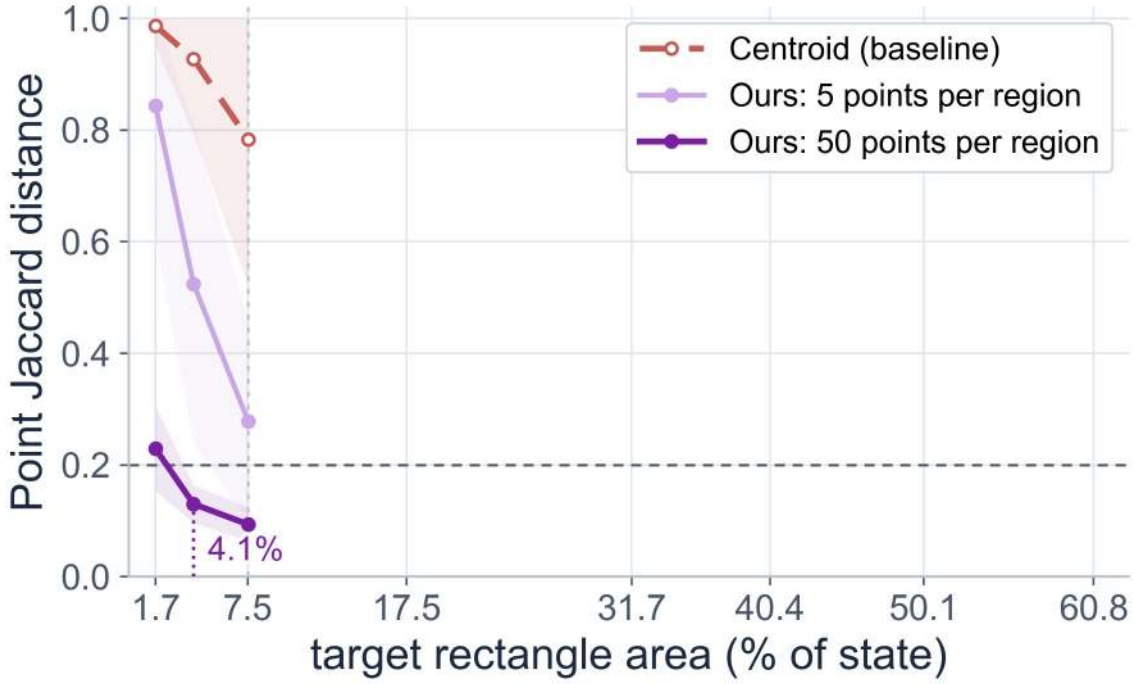




Still small: 50 points per region find it; the centroid does not

Georgia, 159 counties · p − q = 0.4 · 80 trials per size

Georgia: we find regions 15× smaller

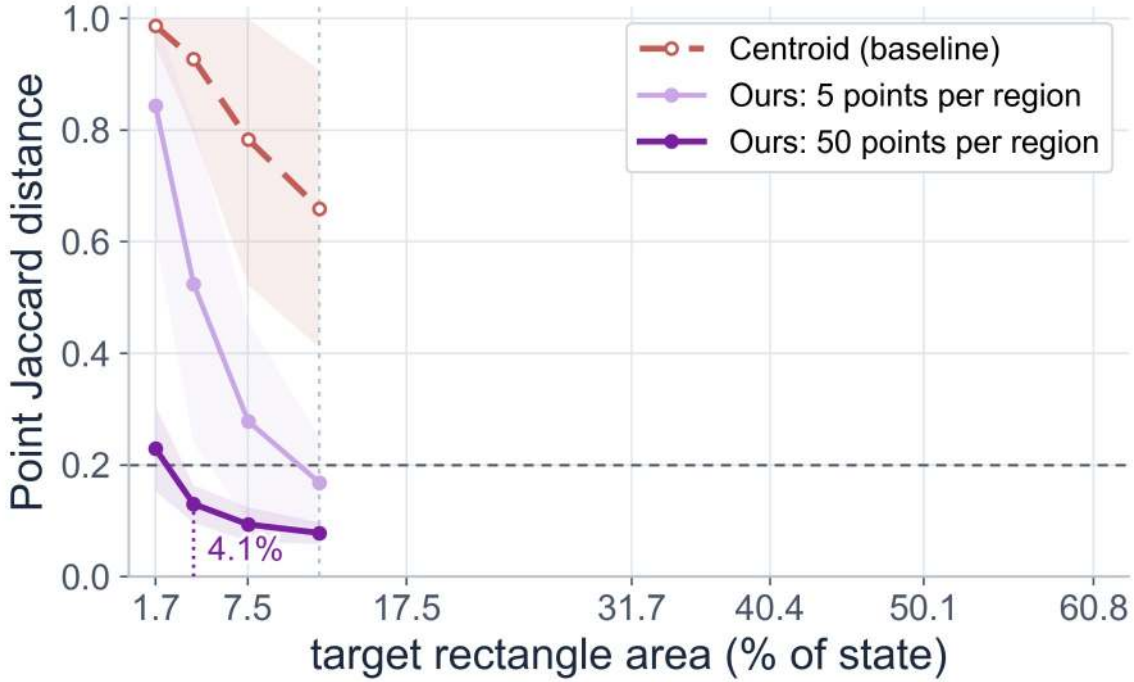


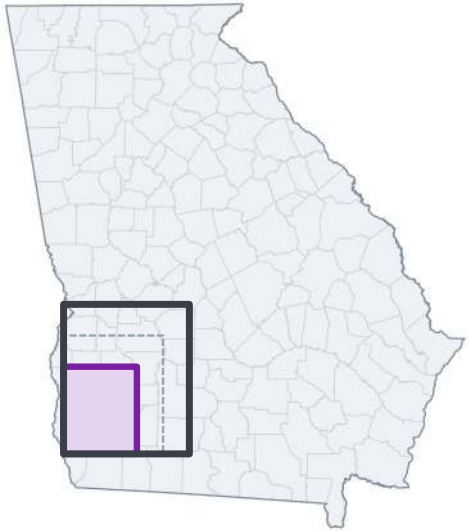


The planted region grows

Georgia, 159 counties · p − q = 0.4 · 80 trials per size

Georgia: we find regions 15× smaller

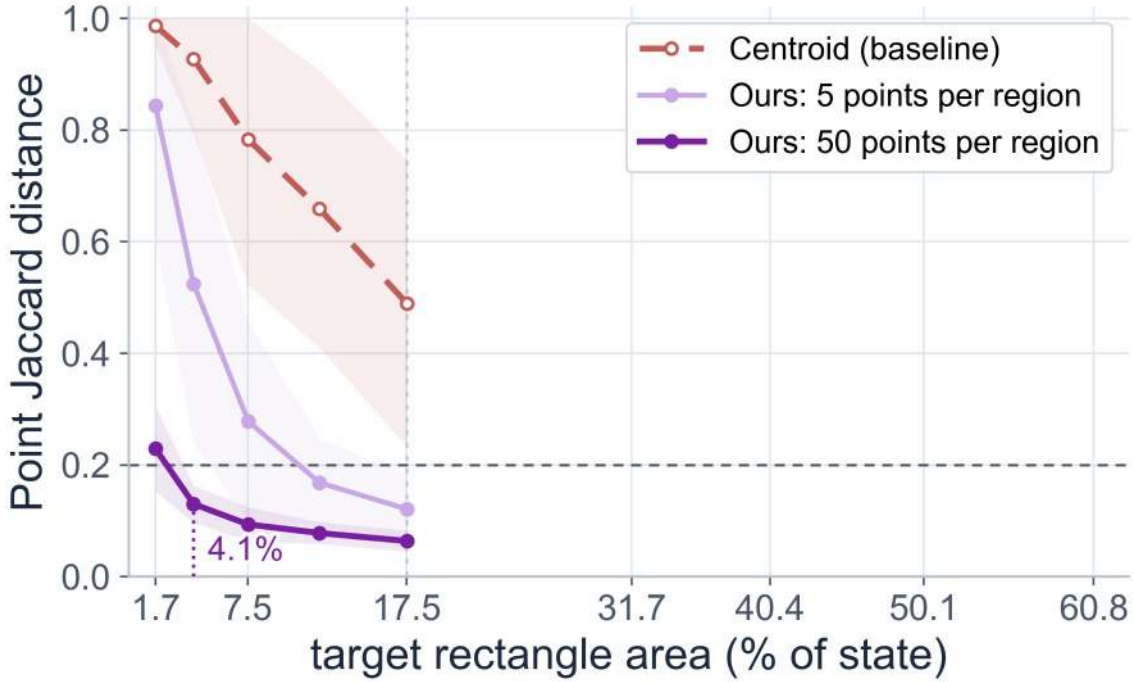


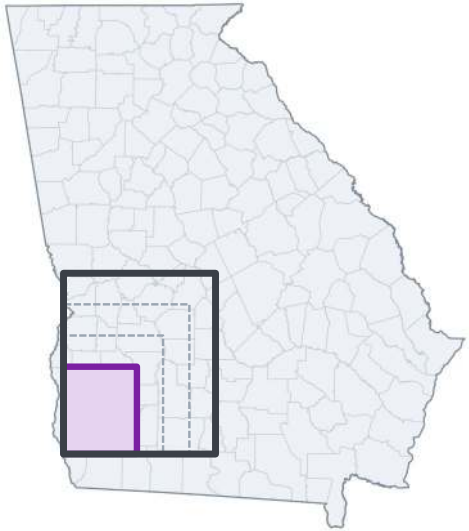


The planted region grows

Georgia, 159 counties · p − q = 0.4 · 80 trials per size

Georgia: we find regions 15× smaller

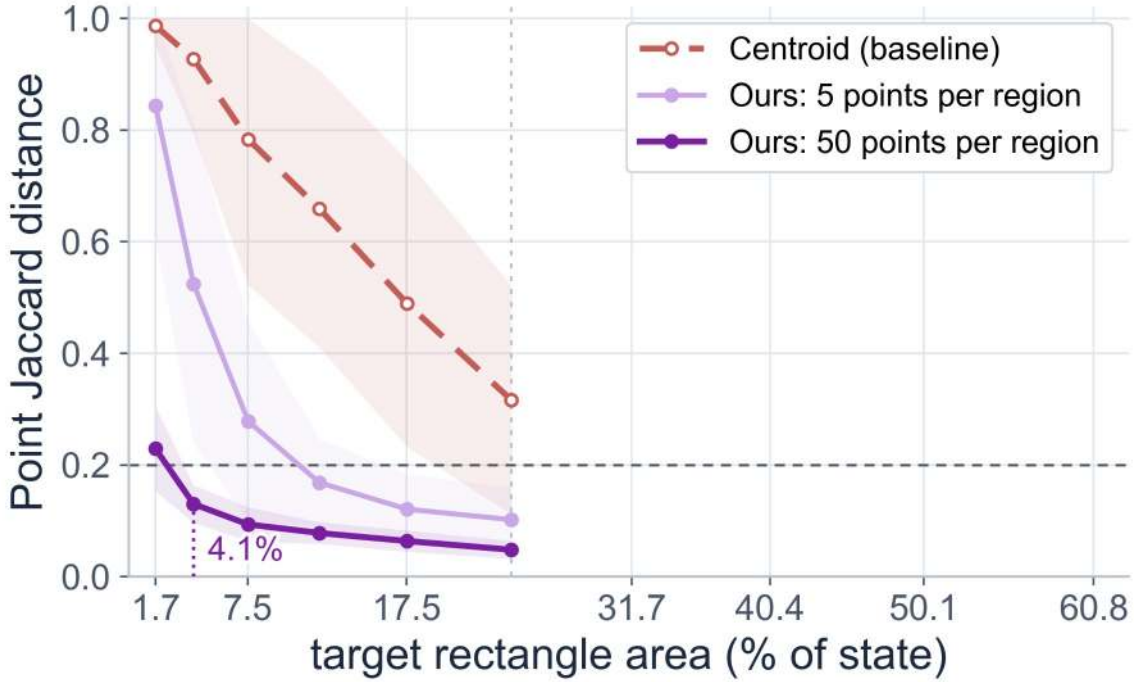


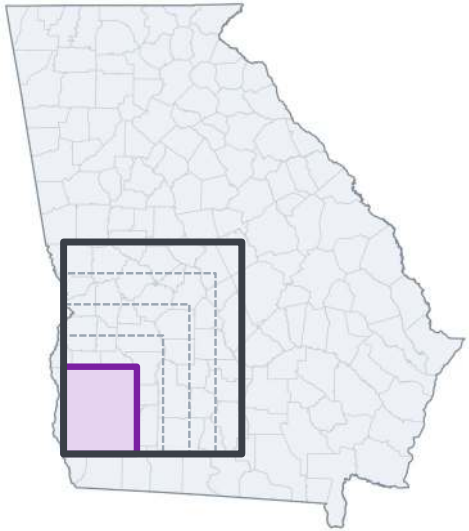


The planted region grows

Georgia, 159 counties · p − q = 0.4 · 80 trials per size

Georgia: we find regions 15× smaller

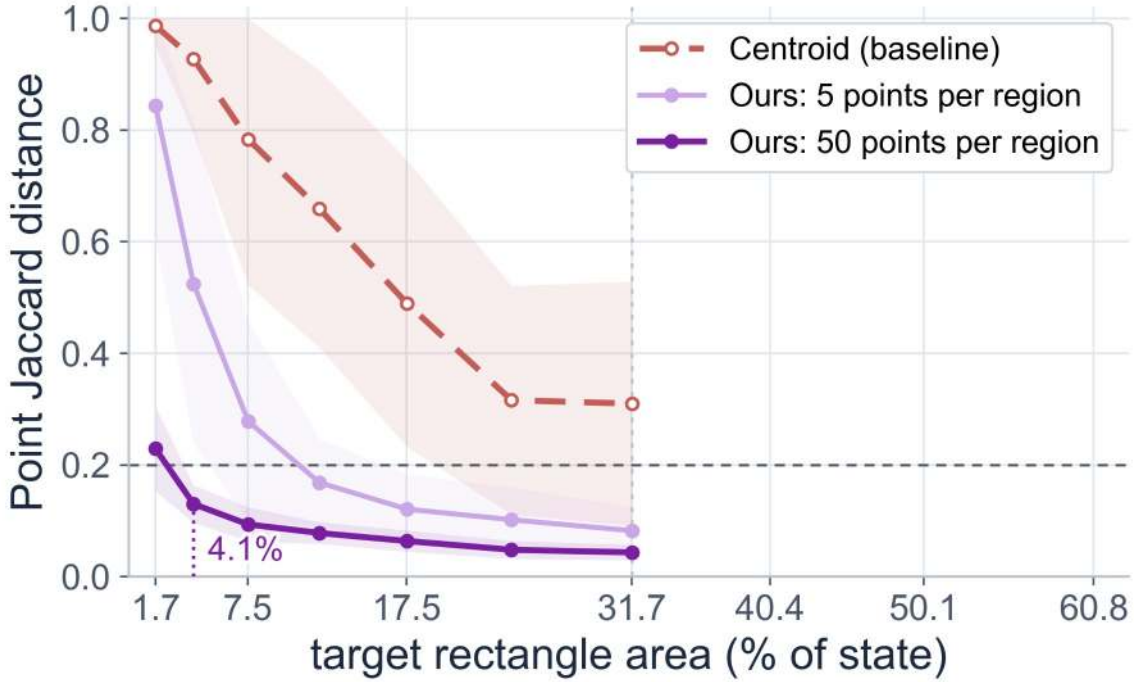


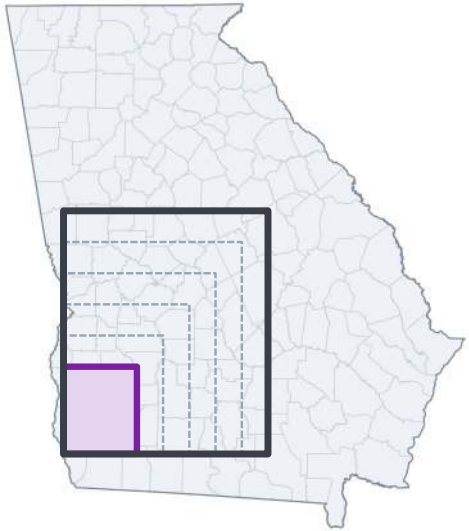


The planted region grows

Georgia, 159 counties · p − q = 0.4 · 80 trials per size

Georgia: we find regions 15× smaller

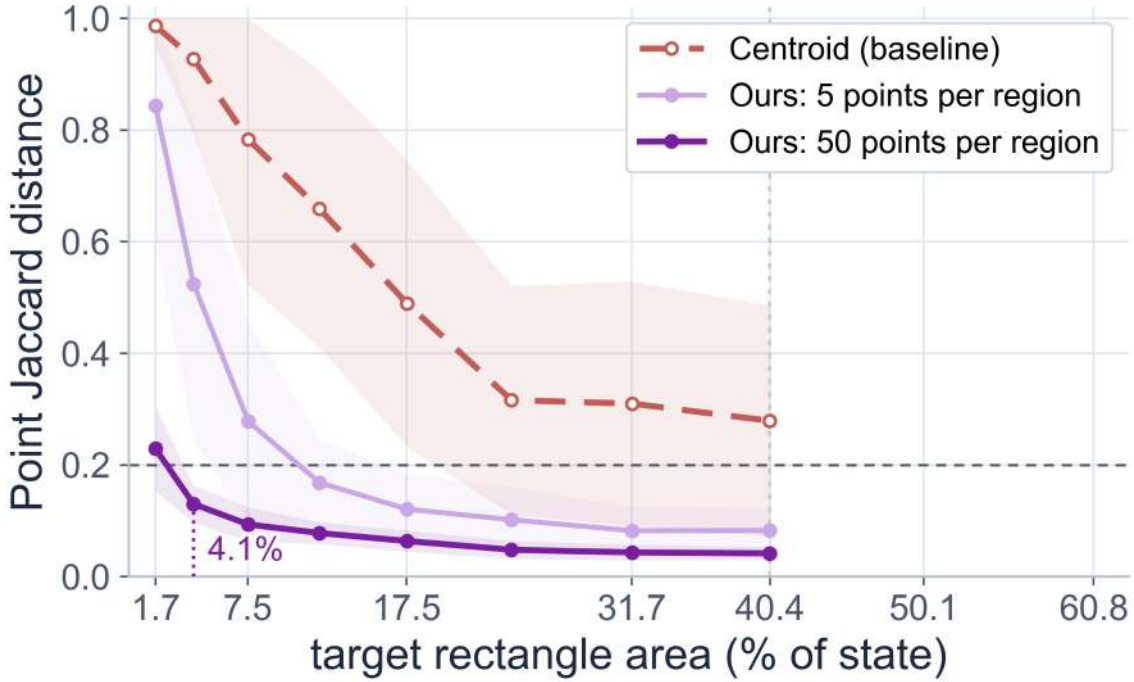


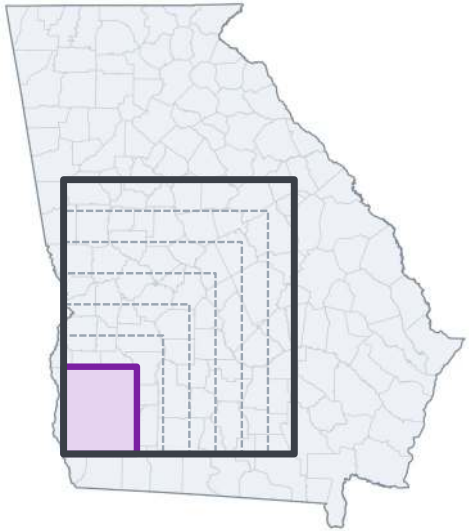


The planted region grows

Georgia, 159 counties · p − q = 0.4 · 80 trials per size

Georgia: we find regions 15× smaller

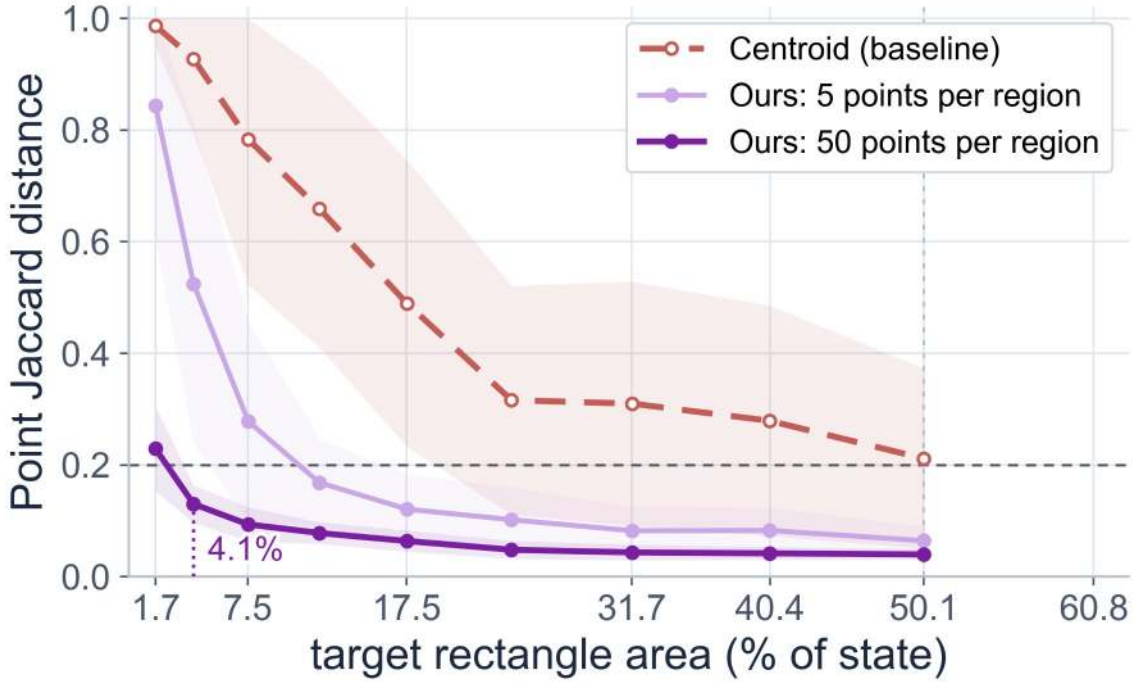


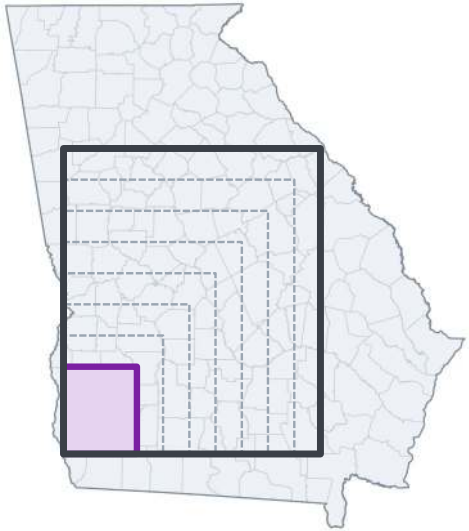


The planted region grows

Georgia, 159 counties · p − q = 0.4 · 80 trials per size

Georgia: we find regions 15× smaller

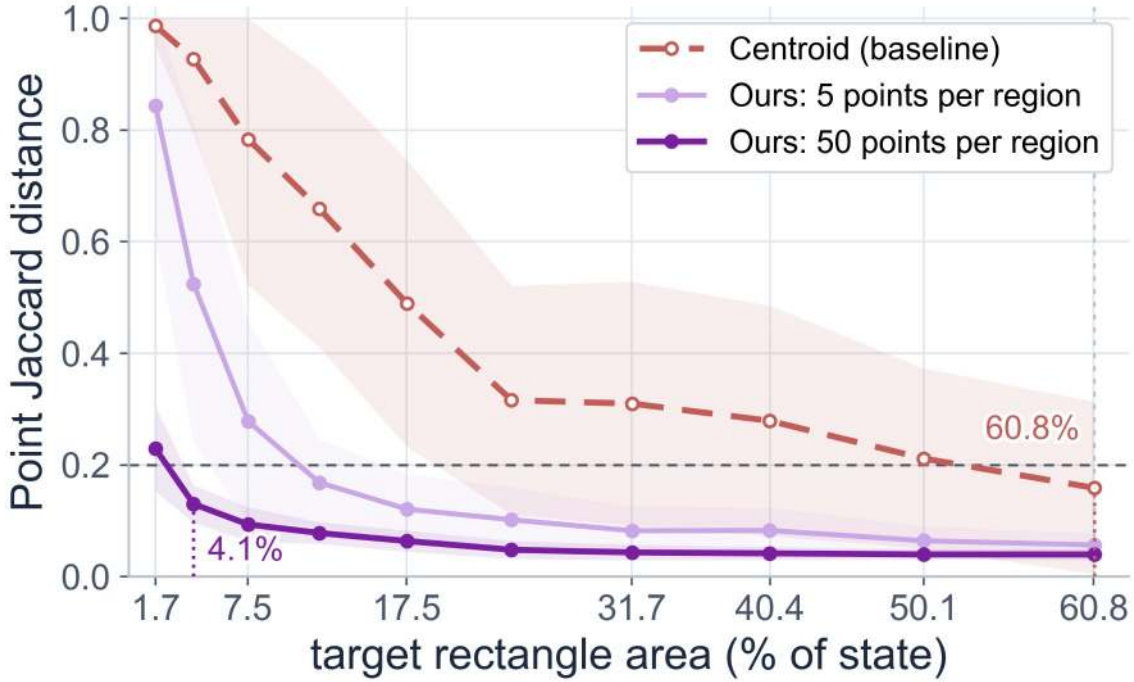


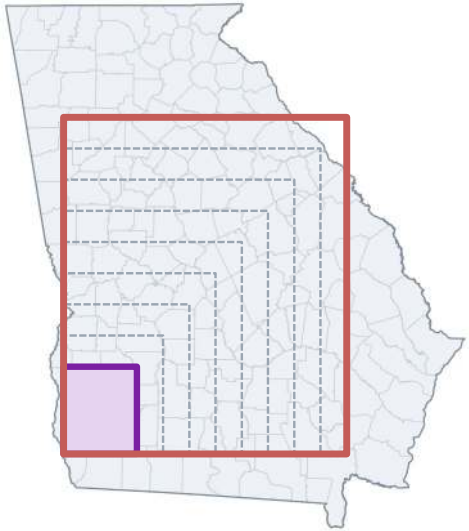


The centroid still fails

Georgia, 159 counties · p − q = 0.4 · 80 trials per size

Georgia: we find regions 15× smaller

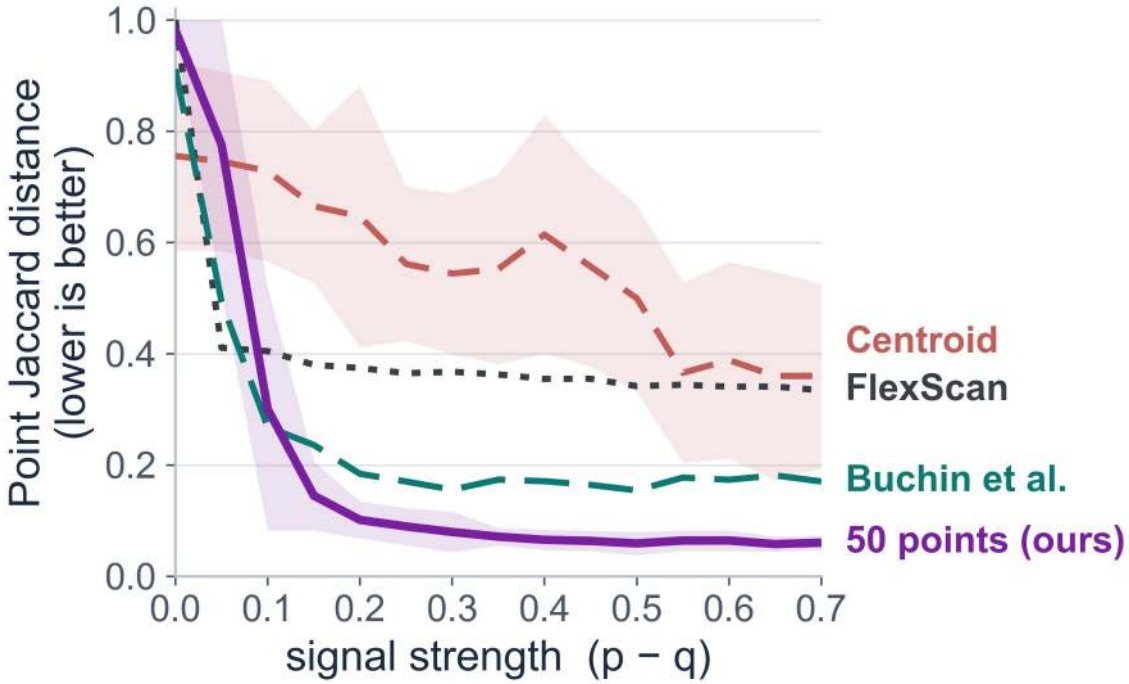




The largest size: the centroid finally succeeds

Georgia, 159 counties · p − q = 0.4 · 80 trials per size

FlexScan and Buchin have built-in limits



FlexScan

- Built-in cap: 15 regions
- The planted region is bigger
- So it stays near 0.35

Buchin et al.

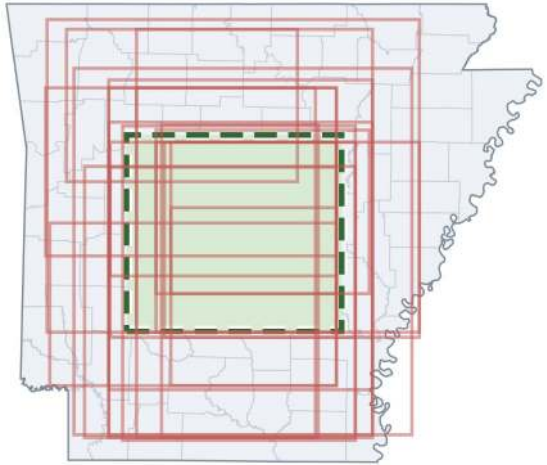
- Needs the window size
- We gave it 9 sizes near the true one
- Still 90× slower than ours

Ours: no cap, no size input.

Arkansas, 75 counties · planted region = 30% of the state · 20 trials per point

Our windows land on the planted region

Centroid



Buchin et al.

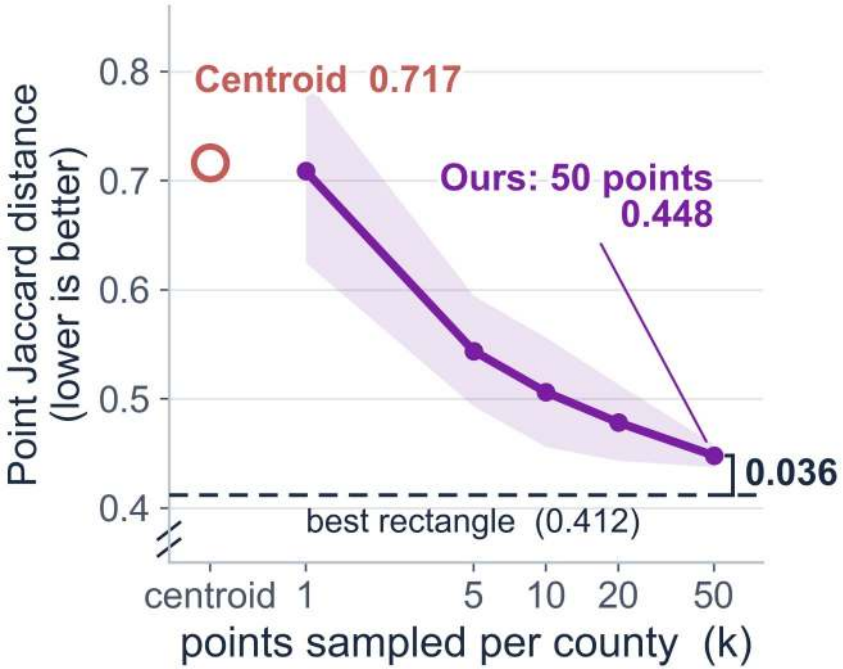


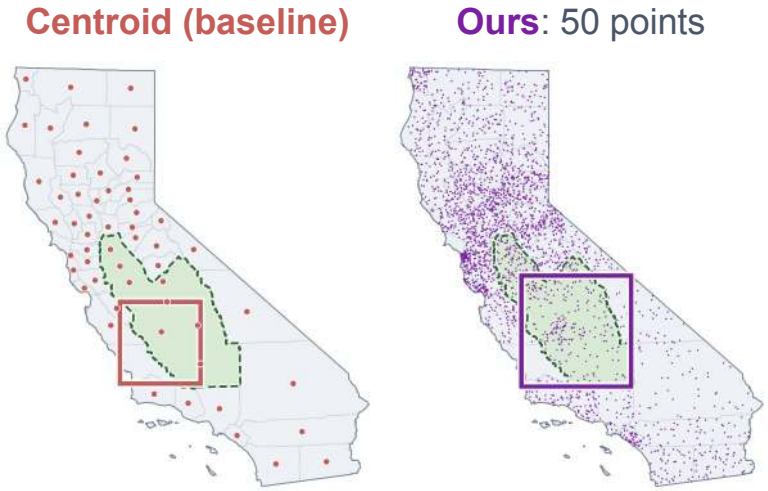
50 points (ours)



Arkansas · p − q = 0.5 · one box per trial, 20 trials · green dashed = planted region

Valley Fever: within 0.036 of the best rectangle





dashed = San Joaquin Valley (the ground truth)

Conclusion: 50 points per region, not one centroid



1 centroid

50 random points

- Better than the centroid on all six maps
- Still fast: 0.33 s on the whole US, with 50× more points
- On real Valley Fever data, the Point Jaccard distance falls from 0.72 to 0.45

Thank you

Foad Namjoo

Seeking a Summer 2027 research internship.

email	foad.namjoo@utah.edu
website	users.cs.utah.edu/~foad27
LinkedIn	linkedin.com/in/foadnamjoo
code	github.com/foadnamjoo/sampling-region-scan







Use cases



Cancer clusters

Kulldorff et al., 2006



Disease outbreaks, such as COVID-19

Desjardins et al., 2020



Emerging public-health threats

Nobles et al., 2022



Street crime

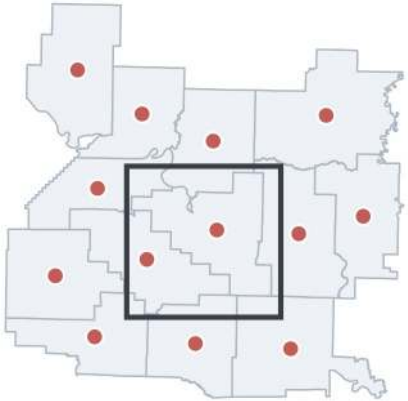
Shiode, 2011

Hotspots of cases: where, and how unusual

backup

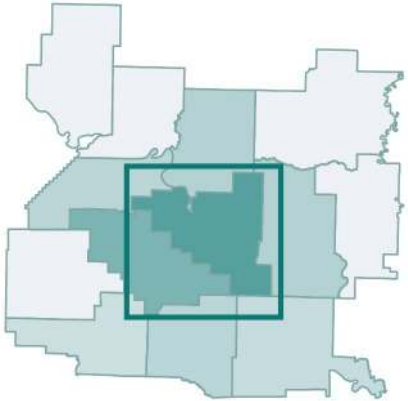
Existing ways to scan regions are lossy, slow, or capped

Centroid



- Fast
- Each region is one dot
- Sees all or nothing

Buchin et al.



- Exact area overlap
- Slow
- Needs the window size

FlexScan

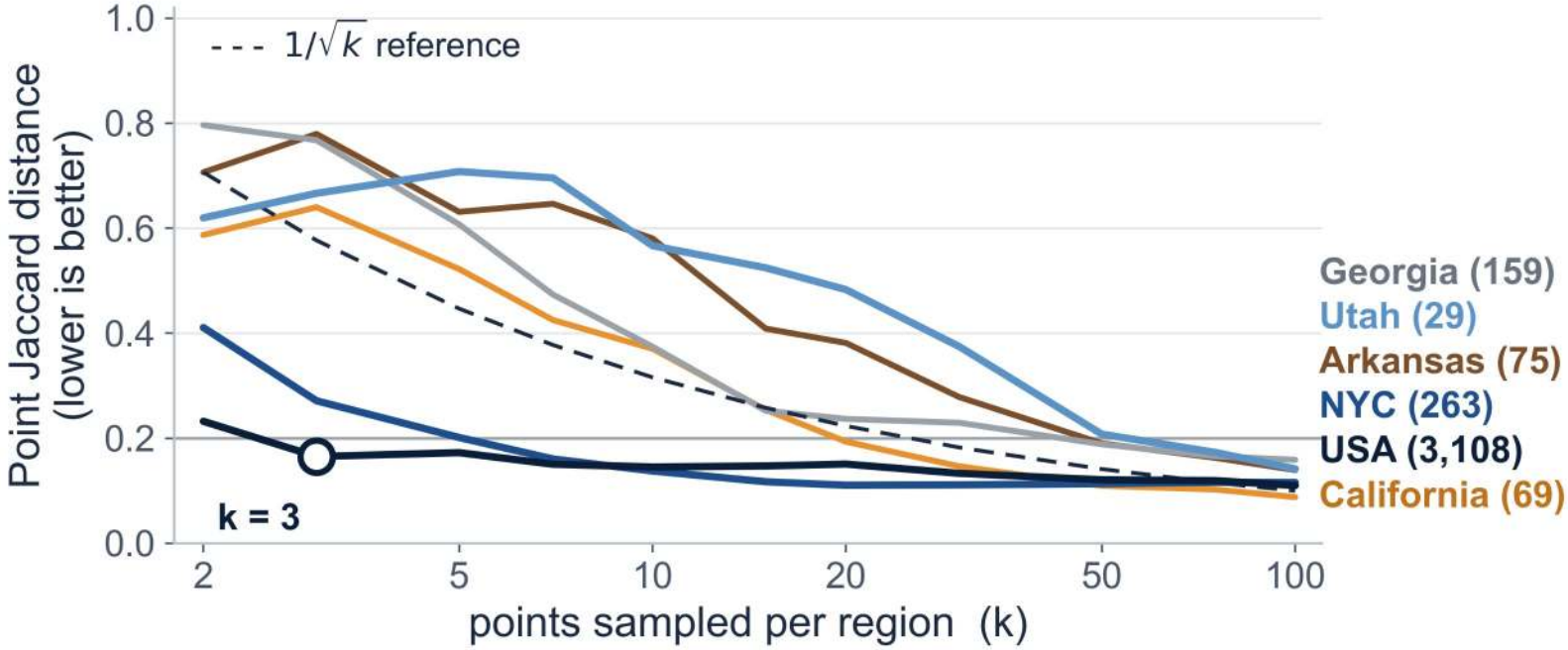


- Whole regions
- Slow
- At most 15 regions

Missing: a fast and accurate way to run point-based scanners on region data

backup

The finest maps need the fewest points: 3 per region for the USA, 50 to 75 for Utah



p − q = 0.15 · 60 trials per k · the theorem is on the next slide

backup

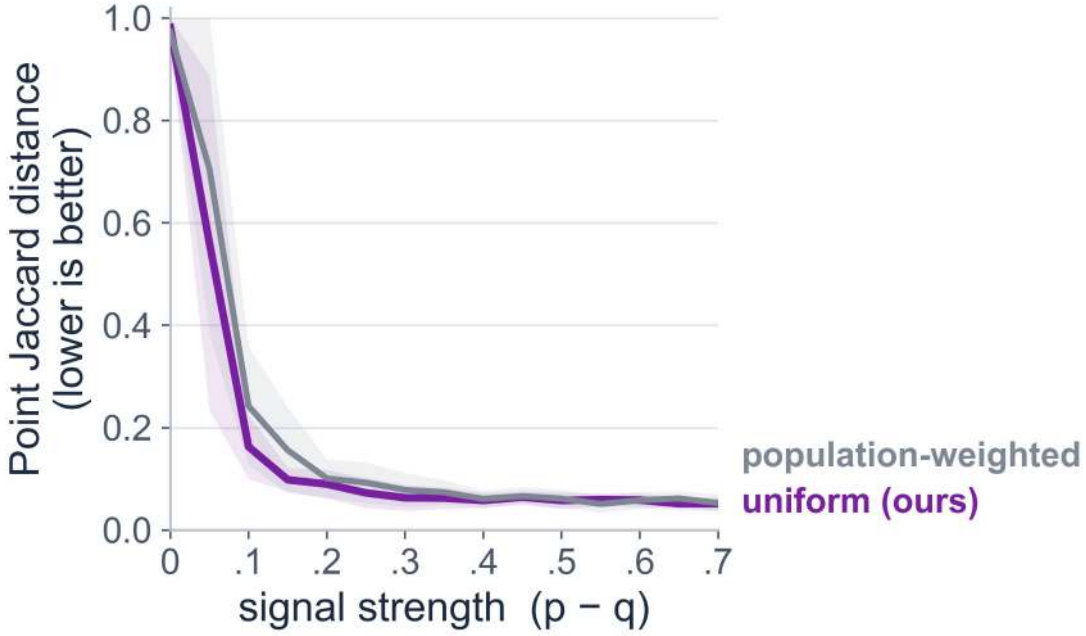
Theorem 3.1 bounds the points needed per region

For an (ε, γ)-stable optimal shape in a κ-balanced set of n regions, sampling **$k = O(\kappa / (n\,\varepsilon^2))$** uniform points per region and running a scan statistic gives a shape within Jaccard distance γ of the optimum, with constant probability.

Proof sketch. Prior sampling results (Matheny et al. 2016, 2018) give score error ε from $O(1/\varepsilon^2)$ samples in total. Spread over n regions, that is $k = O(1/(n\,\varepsilon^2))$ per region; if the largest region holds at most κ times the counts of the smallest, $k = O(\kappa / (n\,\varepsilon^2))$ still gives every region enough samples. (ε, γ)-stability turns a score error ε into a Jaccard error γ.

backup

Weighting the points by population did not help



Georgia, 50 points per county

A capped population-weighted variant (§4.5): dense Atlanta counties hit the cap and rural counties got very few points, which weakened detection of regions spanning low-density areas. The two arms use unequal point budgets.

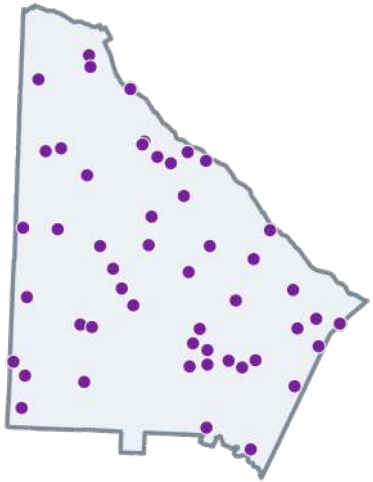
Aggregation has already removed the detail inside each region, so uniform is the honest default.

Georgia, 159 counties · 20 trials per point

backup

Same k in every region: the counts set the weights

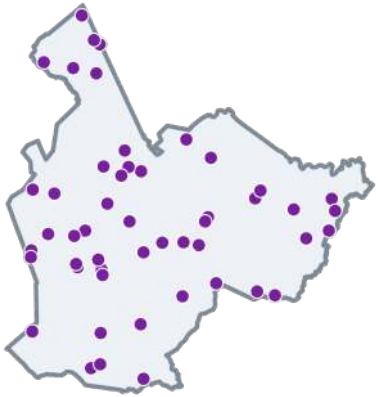
DeKalb (Atlanta)



764,382 people

15,288 per point

Taliaferro (rural)



1,559 people

31 per point

- Each point carries 1/k of its region's cases and people
- So population is in the weights, not in the number of points
- k only sets how finely each region's shape is covered
- Theory: for κ-balanced regions, the same $k = O(\kappa / (n\,\varepsilon^2))$ per region is enough

Georgia, k = 50 points per county · population: 2020 Census

backup